Lecture 1: Basics of probabilities

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Definitions: *Sample Space*, *Events*

Random experiment

By the term ***random experiment*** *we assume any process, which outcome is unknown in advance.*

Sample space

A collection of all possible outcomes (Ω).

Sample point

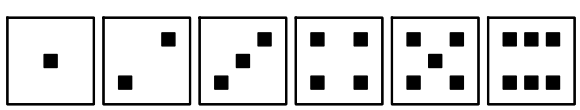
An element $\omega \in \Omega$.

Event

A subset A *of sample points,* $A \subset \Omega$*, for which a statement about an outcome is true.*

Example (Rolling a die)

```
#knitr::include_graphics("./figures/topic01-dice.png")
library(tidydice)
dice <- roll_dice(times = 6)
dice$result <- seq(1, 6)
plot_dice(dice)
```



- ***Random experiment***: *rolling a six-sided die.*
- ***Sample space***: $\Omega = \{1, 2, 3, 4, 5, 6\}$.
- ***Events***:
  - $A =$ "a number is even" $= \{2, 4, 6\}$.
  - $B =$ "a number is divisible by 3" $= \{3, 6\}$.
  - $C =$ "a number is larger than 2" $= \{3, 4, 5, 6\}$.

Operations with events

Sum of events

An event $A + B$ (or $A \cup B$) is called a ***sum of two events***, and it means that *either A or B, or both are true.*

Product of events.

An event $A \cdot B$ (or $A \cap B$) is called a ***product of two events***, and it means that *both A and B are true simultaneously.*

Mutually exclusive events.

Two events A and B are ***Mutually exclusive*** if they cannot be true simultaneously, and $A \cdot B = \emptyset$ (impossible event).

Complement

An event A^c is called a ***complement*** of an event A (or an ***opposite event***):

- $A + A^c = \Omega$ (sample space)
- $A \cdot A^c = \emptyset$ (impossible event)

Probabilities

Probabilities *are numbers assigned to the events and express the chances of events' occurrences.*

1. $\mathrm{P}(\emptyset) = 0$.
2. $\mathrm{P}(\Omega) = 1$.
3. For any event A, $0 \leq \mathrm{P}(A) \leq 1$.

Classical definition

Probability of an event A is defined as

$$\mathrm{P}(A) = \frac{\text{number of sample that belong to the event } A}{\text{total number of sample points in the sample space } \Omega} = \frac{N_A}{N_\Omega}.$$

Remark. The probability defined by the formula above is a proper approach in the situations when each sample point from a sample space has equal chance to occur.

Example (Classical definition of probability)

```r
#knitr::include_graphics("./figures/topic01-dice.png")
library(tidydice)
dice <- roll_dice(times = 6)
dice$result <- seq(1, 6)
plot_dice(dice)
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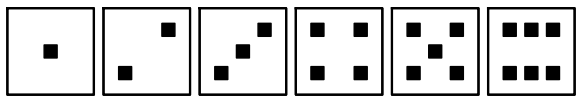


- ***Sample space***: $\Omega = \{1, 2, 3, 4, 5, 6\}$.
- ***Events***:
  - $A =$ "a number is even" $= \{2, 4, 6\} \Rightarrow$

$$\mathrm{P}(A) = \frac{3}{6} = \frac{1}{2}.$$

  - $B =$ "a number is divisible by 3" $= \{3, 6\} \Rightarrow$

$$\mathrm{P}(B) = \frac{2}{6} = \frac{1}{3}.$$

  - $C =$ "a number is larger than 2" $= \{3, 4, 5, 6\} \Rightarrow$

$$\mathrm{P}(C) = \frac{4}{6} = \frac{2}{3}.$$

Rules for probabilities

1. For any two events A and B,

$$\mathrm{P}(A + B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cdot B).$$

2. Let $A_1, A_2, \ldots$ be an infinite sequence of ***mutually excluding*** (i.e. $A_i \cdot A_j = \emptyset$, if. $i \neq j$) events. Then

$$\mathrm{P}(\text{"At least one of } A_i \text{ is true"}) = \mathrm{P}\left(\sum_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathrm{P}(A_i).$$

3. Conditional probability

$$\mathrm{P}(A|B) = \frac{\mathrm{P}(A \cdot B)}{\mathrm{P}(B)},$$

   the probability that A is true when we ***know*** that some event B is true.

Independence of events

Two events $A, B \subset \Omega$ are called ***independent*** *if*

$$\mathrm{P}(A \cdot B) = \mathrm{P}(A) \cdot \mathrm{P}(B).$$

Two events A and B are ***dependent*** *if they are not independent, i.e.*

$$\mathrm{P}(A \cdot B) \neq \mathrm{P}(A) \cdot \mathrm{P}(B).$$

For two ***dependent*** events A and B,

$$\mathrm{P}(A \cdot B) = \mathrm{P}(A) \cdot \mathrm{P}(B|A) = \mathrm{P}(B) \cdot \mathrm{P}(A|B).$$

Law of total probability

Partition of sample Space

A collection of events $A_1, A_2, \ldots, A_n$ is called a ***partition*** *of a sample space Ω if*

(i) *The events are mutually excluding, i.e.*

$$A_i \cdot A_j = \emptyset, \ i \neq j.$$

(ii) *The collection is exhausive, i.e.*

$$A_1 + A_2 + \ldots + A_n = \Omega,$$

that is, at least one of the events A_i occurs.

Law of total probability

Let $A_1, A_2, \ldots, A_n$ be a partition of a sample space Ω. Then, for any event B,

$$\mathrm{P}(B) = \mathrm{P}(B|A_1)\,\mathrm{P}(A_1) + \mathrm{P}(B|A_2)\,\mathrm{P}(A_2) + \ldots + \mathrm{P}(B|A_n)\,\mathrm{P}(A_n).$$

Example (Law of total probability)

- An event B is "Experiencing an adverse event (AE) for a patient in the comparative clinical trial with two experimental (E_1, E_2) and one control (C) treatment."
- From the previous studies, it is known that
  - 1 out of 50 patients experienced AE on treatment E_1.
  - 1 out of 25 patients experienced AE on tretament E_2.
  - 1 out of 10 patients experienced AE on tretament C.
- Patients are randomzied to treatments with probabilities

$$p_{E_1} = 0.5, \ p_{E_2} = 0.3 \ p_C = 0.2.$$

Q: *What is the probability of AE for a random patient?*

Solution:

- Let us consider the followinf partition of a sample space:

  - A_1 = "A patient is randomzied to treatment E_1".
  - A_2 = "A patient is randomzied to treatment E_2".
  - A_3 = "A patient is randomzied to treatment C".

- A_1, A_2, A_3 are mutually excluding but at least one of them is true.
- Then we can define:

$$\begin{array}{ll} \mathrm{P}\left(B|A_1\right) = \frac{1}{50} = 0.02 & \mathrm{P}\left(A_1\right) = p_{E_1} = 0.5 \\ \mathrm{P}\left(B|A_2\right) = \frac{1}{25} = 0.04 & \mathrm{P}\left(A_2\right) = p_{E_2} = 0.3 \\ \mathrm{P}\left(B|A_3\right) = \frac{1}{10} = 0.10 & \mathrm{P}\left(A_3\right) = p_C = 0.2 \end{array}$$

$$P(B) = 0.02 \cdot 0.5 + 0.04 \cdot 0.3 + 0.10 \cdot 0.2 = 0.042.$$

Bayes' formula

Bayes' theorem

Let $A_1, \ldots, A_n$ be a partition of a sample space Ω, and B is an event with $\mathrm{P}\left(B\right) > 0$.

- $\mathrm{P}\left(A_i\right)$, $i = 1, \ldots, n$, is a **prior probability**.
- $L(A_i) = \mathrm{P}\left(B|A_i\right)$, $i = 1, \ldots, n$, is a **likelihood** (measures how likely the observed event B under the alternative A_i).

Then a **posterior probability**

$$\mathrm{P}\left(A_i|B\right) = \frac{\mathrm{P}(A_i \cap B)}{\mathrm{P}(B)} = \frac{\mathrm{P}(A_i)\mathrm{P}(B|A_i)}{\mathrm{P}(B)}.$$

Name due to *Thomas Bayes* (1702-1762)

Bayes' formula

Recall that, due to the law of total probability,

$$\mathrm{P}(B) = \mathrm{P}(B|A_1)\mathrm{P}(A_1) + \ldots + \mathrm{P}(B|A_n)\mathrm{P}(A_n).$$

Then, the Bayes' formula can be rewritten as

$$\mathrm{P}\left(A_i|B\right) = c\mathrm{P}(A_i)\mathrm{P}(B|A_i)\left(c = \frac{1}{\mathrm{P}(B)}\right).$$

In practical computations, all terms $\mathrm{P}(A_i)\mathrm{P}(B|A_i)$, $i = 1, \ldots, n$, are evauated, then c is derived.

Odds and subjective probabilities

```
knitr::include_graphics("./figures/topic01-bitcoin.png")
```

Two events associated with the experiment “tossing a coin”:

- A_1 = "A coin shows a "head""
- A_2 = "A coin shows a "tail""

If a coin is ***fair*** than the ***odds*** for A_1 and A_2 is $1:1$.

In the more general case, the ***odds*** for two events A_1 and A_2 are defined as any posisitve numbers q_1, q_2 such that

$$\frac{q_1}{q_2} = \frac{\mathrm{P}(A_1)}{\mathrm{P}(A_2)}.$$

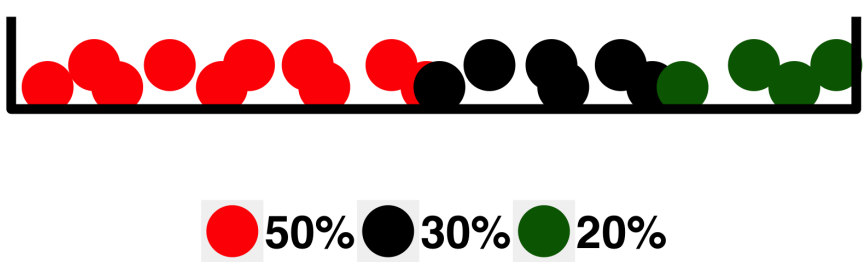
- Probabilities ***are known*** $\Rightarrow$ odds ***can always be found***.
- Odds are known $\not\Rightarrow$ probabilities ***can always be found***. (a example with just two outcomes considered)

Let $A_1, A_2, \ldots, A_n$ be a partition of Ω, having odds q_i, i.e. for any i, j, $\frac{\mathrm{P}(A_j)}{\mathrm{P}(A_i)} = \frac{q_j}{q_i}$. Then

$$\mathrm{P}(A_i) = \frac{q_i}{q_1 + q_2 + \ldots + q_n}, \; i = 1, 2, \ldots, n.$$

Example (Odds)

```
knitr::include_graphics("./figures/topic01-urn-with-balls.png")
```



- An experiment is to draw a ball from the urn.
  - A_1 = "a ball is red"
  - A_2 = "a ball is black"
  - A_3 = "a ball is green"
- A_1, A_2, A_3 forms a partition.
- Odds: $q_1 : q_2 : q_3 = 5 : 3 : 2$.
  - $\mathrm{P}(A_1) = \frac{5}{5+3+2} = 0.5$, $\mathrm{P}(A_2) = 0.3$, $\mathrm{P}(A_3) = 0.2$.

Bayes' formula for odds

- Let $A_1, A_2, \ldots, A_n$ be a partition of Ω with the *a priori* odds $q_1^{\text{prior}}, q_2^{\text{prior}}, \ldots, q_n^{\text{prior}}$.
- The *a posteriory* odds are any positive numbers $q_1^{\text{post}}, q_2^{\text{post}}, \ldots, q_n^{\text{post}}$ such that

$$\frac{\mathrm{P}(A_i|B)}{\mathrm{P}(A_j|B)} = \frac{q_i^{\text{post}}}{q_j^{\text{post}}}.$$

- By applying Bayes'formula, we have

$$\frac{q_i^{\text{post}}}{q_j^{\text{post}}} = \frac{\mathrm{P}(A_i|B)}{\mathrm{P}(A_j|B)} = \frac{\mathrm{P}(B|A_i)\mathrm{P}(A_i)}{\mathrm{P}(B)} \cdot \frac{1}{\frac{\mathrm{P}(B|A_j)\mathrm{P}(A_j)}{\mathrm{P}(B)}} = \frac{\mathrm{P}(B|A_i)}{\mathrm{P}(B|A_j)} \cdot \frac{q_i^{\text{prior}}}{q_j^{\text{prior}}}$$

  which implies that

$$q_i^{\text{post}} = c\mathrm{P}(B|A_i)q_i^{\text{prior}},\ i = 1, 2, \ldots, n,$$

  where c is any positive number.

Example (Bayes' formula for odds)

A newspaper reports that the first case of a susbected "mad cow"[1] is found. "Suspected" means that a laboratory test for the illness gave positive result.

Since this information can influence shopping habbits, a preliminary risk analysis is desired.

The most important information is the probability that a cow, *positively tested for BSE*, is ***really infected***.

Mathematical formulation of the problem:

- $A =$ "A cow is BSE infected"
- A partition:
  - $A_1 = A =$ "A cow is BSE infected"
  - $A_2 = A^c =$ "A cow is NOT BSE infected"
- $B =$ "A cow is positively tested for BSE"

The *posterior* odds, $q_1^{\text{post}}, q_2^{\text{post}}$, for A_1 and A_2 can be computed using Bayes' formula for odds, if *prior* odds, $q_1^{\text{prior}}, q_2^{\text{prior}}$, and the conditional probabilities $\mathrm{P}(B|A_1)$ and $\mathrm{P}(B|A_2)$ are known:

$$q_i^{\text{post}} = \mathrm{P}(B|A_i)q_i^{\text{prior}},\ i = 1, 2.$$

Selection of prior odds

Assume that the following information is available (say from the previous experience, or from the *Internet*):

- The frequency of infected cows that pass the test, i.e. are ***not detected***, is 1 per 100 $\Rightarrow P(B|A_1) = 0.99$.
- A healthy cow is suspected for having BSE in 1 per 1000 cases $\Rightarrow P(B|A_2) = 0.001$.
- $q_1^{\text{prior}} : q_2^{\text{prior}} = 1 : 1$.

Then, $q_1^{\text{post}} = 0.99 \cdot 1 = 0.99,\ q_2^{\text{post}} = 0.001 \cdot 1 = 0.001$, and

$$q_1^{\text{post}} : q_2^{\text{post}} = 990 : 1.$$

[1] BSE (Bovine Spongiform Encephalopathy) infected cow